Sai Kiran <ai24mtech02003@iith.ac.in>

Notification of Acceptance: IEEE ICIP 2026 - #1874

icip2026 <notifications@exordo.com> Thu, Apr 30, 2026 at 9:01 PM
Reply-To: Ex Ordo Support Team <support@exordo.com>
To: ai24mtech02003@iith.ac.in

13 - 17 September 2026

IEEE International Conference on Image Processing 2026

https://icip2026.exordo.com

Dear Sai Kiran Kocherla,

Following careful review by the external reviewers, area chairs and technical program chairs, I am pleased to inform you that your paper ID #1874 entitled “SIGMA-Based RGB-Hyperspectral Fusion for Semantic Segmentation in Autonomous Driving” has been accepted for the IEEE ICIP 2026 conference. Congratulations!

Final Submission and Next Steps
As you prepare the camera-ready paper, please ensure you incorporate the recommendations from the reviewers and chairs, which you can find appended at the bottom of this message. You may find additional comments by logging into your IEEE ICIP 2026 dashboard at https://icip2026.exordo.com/

Final Submission & Copyright
You will receive a separate email containing the timeline, deadlines, and instructions for uploading your camera-ready paper and completing the IEEE Copyright Form (eCF).

Action Required
Please await this follow-up email before finalising your submission in the system.

Presentation format
Details regarding the presentation format (e.g., oral or poster) will be communicated later on, as the conference program is finalized**.**

Travel support and SPS membership
The Signal Processing Society offers active members with an accepted paper for this conference the possibility to apply for travel support to help alleviate travel costs to the conference to present your paper. Details, including eligibility criteria, application deadlines, and instructions on how to apply, are available on the SPS Travel Grants webpage. If you are not yet an IEEE SPS Member, please consider joining to be eligible for travel grants and a reduced registration fee.

Thank you for your contribution to IEEE ICIP 2026 and congratulations on the acceptance!

We look forward to meeting you in Tampere!

Sincerely,
Alessandro Foi
Lead Technical Program Chair, IEEE ICIP 202

_____

📎 Submission

SIGMA-Based RGB-Hyperspectral Fusion for Semantic Segmentation in Autonomous Driving

- Sai Kiran Kocherla (Indian Institute of Technology Hyderabad)
- Srinivas Aditya Abbaraju (Indian Institute of Technology Hyderabad)
- Adduru U G Sankararao (Indian Institute of Information Technology Sricity)
- Duswanth Reddy A (TIHAN-IIT Hyderabad)
- Rajalakshmi P (Indian Institute of Technology Hyderabad)

Abstract

Autonomous driving semantic segmentation has been mostly based on RGB imagery, which can be easily confused by similar materials that have similar appearance even in adverse lighting conditions. Hyperspectral imaging HSI is capable of providing enhanced spectral information that can be useful in adding to RGB information, and yet fusing these heterogeneous modalities is difficult because of dimensional differences and representation. In this paper, we apply the SIGMA multi-modal fusion framework to the problem of RGB-HSI semantic segmentation by adding a simple adaptation module that encodes both modalities into a common embedding space. The adapter allows stable learning of shared-backbone and retains modality-specific properties. We test the suggested methodology on several publicly available hyperspectral driving datasets and present similar gains in mean Intersection-over-Union and pixel accuracy compared to the unimodal and naive fusion baselines. The significance of the design of the adapters is confirmed by the extensive ablation research that proves that hyperspectral information is effective in material-conscious perception in autonomous driving.

Keywords

Semantic Segmentation, Autonomous Driving, Hyperspectral Imaging, SIGMA, multimodal

Topics

- Machine learning for 3D image and video processing

- Color and multispectral imaging
- Autonomous systems

Additional Info

Paper PUBLISH-READY version:	Verified_publish_ready_26_ICIP_MAMBA_2026049350.pdf
Paper DOUBLE BLIND ANONYMISED version:	Anonymised-Filename-1874-20260212111729.pdf
Supplemental Materials File 1 (Optional)::	—
Supplemental Materials File 2 (Optional)::	—
IEEE Event Terms and Conditions::	Yes
IEEE Privacy Policy::	Yes
IEEE ICIP2026 Author Rebuttal::	We sincerely thank all reviewers for their constructive feedback. Our work investigates RGB-HSI fusion within a shared-backbone framework, and we clarify key aspects of motivation, design and evaluation below. Reviewer 1: Comment 1: Missing comparison with RGB SOTA methods Our objective is not to propose a new segmentation architecture competing with RGB SOTA methods, but to isolate the contribution of hyperspectral information. We adopt a shared architecture(SIGMA) and compare RGB-only, HSI-only, and RGB+HSI, eliminating architectural bias. Importantly, Table 1 already includes strong RGB baselines across datasets (Pg 4), where our method achieves competitive or superior performance, demonstrating the benefit of spectral fusion. Comment 2: HSI-only

underperforming RGB
While hyperspectral data is theoretically richer, in practice it is affected by lower spatial resolution,sensor noise and sparse/heterogeneous annotations(Sec. 1-2), explaining weaker HSI-only performance(Table 2,mIoU 15.47). Importantly, RGB+HSI significantly improves performance(mIoU 80.42), showing that spectral information is complementary,not redundant and most effective when combined with RGB spatial structure.

Comment 3: Motivation for hyperspectral data
The HSI captures material-specific spectral signatures beyond visible RGB,enabling discrimination of visually similar materials(e.g.,asphalt vs concrete) that are ambiguous in RGB.This is discussed in Sec 1(Pg 1) and supported by qualitative results(Fig. 2, Pg 5), where the proposed model better separates such regions compared to RGB baselines.

Comment 4: Clarity of datasets and metrics
The datasets (HyKo2-VIS, HyKo2-NIR, HSI-DriveV2, HSI-Road) and evaluation metrics are described in Sec 4.1 (Pg 4).We agree that clarity can be improved and will make these definitions more explicit.

Comment 5: Simplicity / ICIP relevance
We would like to clarify that the novelty lies in adapting shared-backbone multimodal fusion to

hyperspectral data, which is inherently high-dimensional and heterogeneous.Unlike prior RGB-Depth or RGB-Thermal; RGB-HSI introduced, addressing spectral dimensionality mismatch, modality heterogeneity and sparse annotations(Sec 3.3, Pg 3). Our proposed adapter module provides a lightweight yet effective solution to align modalities into a shared embedding space,enabling stable fusion within the SIGMA framework.We believe this contributes to image processing research by enabling effective use of spectral information in real-world perception tasks.

Reviewer 2:
Comments :Equation clarification and reference update
The variable y_t in Eq.(1) represents the output at spatial position t, derived from the hidden state h_t. We will explicitly define this in the revised version and update the SIGMA reference to WACV 2025.

Reviewer 3:
Comment 1 : Novelty clarification
Our contributions are: 1.extending SIGMA to RGB-HSI fusion,which is non-trivial due to high spectral dimensionality. 2.Introducing a learnable adapter module that aligns heterogeneous modalities into a shared embedding space. 3.demonstrating consistent improvements across four hyperspectral driving datasets.Unlike prior work,our study focuses on material-aware segmentation in autonomous driving.

	Comment 2 : mIoU vs Precision/F1 discrepancy The discrepancy between mIoU and precision/F1 arises due to class imbalance, which is significant in hyperspectral datasets. mIoU reflects region-level overlap, while precision/F1 are sensitive to false positives in minority classes. As noted in Sec. 4.2, mIoU is a more reliable indicator in this setting; we will clarify this further. Comment 3 : VSSM explanation and writing improvements We agree that the explanation of VSSM can be expanded.We will provide additional intuition on how state-space modeling captures long-range dependencies with linear complexity and why it is suitable for dense prediction tasks. We will carefully revise the manuscript to remove redundancies and improve the clarity.
IEEE ICIP2026 Author Rebuttal file upload::	—

Other Properties

Track:	Regular Papers, Special Session Papers
Presentation Format:	Regular Papers and Special Session Papers
Submission ID:	1874

📘 Reviewer Comments and Technical Program Chair Comments

Reviewer 3

Author Feedback

This paper presents a very interesting approach to a real problem in autonomous driving, which is the difficulty of identifying materials that look similar in traditional RGB images. By combining RGB data with Hyperspectral Imaging (HSI) through the SIGMA framework, the authors show that it is possible to distinguish between

surfaces like asphalt and concrete that often confuse standard sensors.. I believe the proposed Adapter Module is a smart and practical solution to align these two very different types of data into a common space without losing the rich spectral information. It is also great to see the model tested on four different datasets, as this proves that the fusion strategy is stable across different scenarios. However, I have some concerns regarding the novelty of the work, as it feels mostly like an adaptation of an existing architecture; therefore, I think the authors should better highlight what makes their specific contribution unique. Additionally, some of the experimental results are quite confusing and need a much deeper explanation. For instance, in the HyKo2-NIR dataset, the model achieves a very high mIoU of 85.59%, but the Precision and F1-score are significantly lower than other benchmark methods, which is a gap that isn't fully discussed. I also felt that the explanation of the Vision State Space Model (VSSM) was a bit brief and could be expanded to help readers understand why it is superior to traditional methods for this task. Finally, the writing in the introduction and methodology is somewhat repetitive, and a careful proofreading would help to fix minor typos and improve the overall professional feel of the paper. If the authors can clarify these technical points and provide a stronger analysis of the data, this will be a very solid and useful paper for the conference.

Reviewer 1

Author Feedback

This paper proposes and evaluates a methodology for semantic segmentation in autonomous driving, with the primary contribution being the incorporation of spectral imaging in addition to standard RGB data. The approach builds on an existing semantic segmentation framework (Sigma) and applies a conventional learning-based fusion strategy to combine inputs of differing dimensionalities. Experimental results are presented on two relatively small autonomous-driving datasets that include spectral measurements.

However, the experimental section lacks clarity and completeness. The datasets and evaluation metrics are not adequately defined, and several essential citations and explanations are missing (metrics, datasets). Given the extensive prior work on semantic segmentation for autonomous driving, it is surprising that no comparisons are provided against established state-of-the-art RGB-based methods. Without such baselines, it is difficult to assess whether the inclusion of spectral information offers any genuine benefit.

Additionally, the ablation study reveals that the spectrum-only model performs substantially worse than the RGB-only model. This is noteworthy because spectral data should, in principle, contain all information present in RGB channels. This discrepancy warrants deeper investigation. For instance, converting the spectral data to RGB and evaluating the resulting performance could help clarify this issue, but such analysis is currently absent and would require a detailed explanation.

Several parts of the paper are discussed only superficially. The motivation for using spectral information, in particular, needs a more rigorous justification supported by examples that demonstrate clear advantages. Human vision operates with only RGB channels and is highly effective for tasks such as driving and scene understanding, which further underscores the need for stronger motivation.

Overall, the proposed methodology is relatively simple, and the experimental results raise more questions than they resolve. The presented evidence does not convincingly support the claimed benefits of spectral information for semantic segmentation in autonomous driving. Furthermore, the topic may be better suited to the computer vision community. An image-processing audience would expect a more thorough, technically grounded analysis of the spectral data itself rather than high-level experiments without substantial insight.

Minor: the Sigma reference should be updated as the work is published in WACV. Moreover, the reference number in Figure 1 is pointing to a wrong reference (should be [6] instead of [9]).

Reviewer 2

Author Feedback

In this paper an adapter-based SIGMA framework for semantic segmentation in autonomous driving which fuses RGB and hyperspectral information is proposed. The suggested SIGMA-RGBHSI model delivers competitive or sota results and provides a base for its future application in autonomous driving systems.

Some comments:
- In Eq.(1) y_t has not been defined.
- ArXiv reference 6 could be replaced by this one

@inproceedings{wan2025sigma,
title={Sigma: Siamese mamba network for multi-modal semantic segmentation},
author={Wan, Zifu and Zhang, Pingping and Wang, Yuhao and Yong, Silong and Stepputtis, Simon and Sycara, Katia and Xie, Yaqi},
booktitle={2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)},
pages={1734--1744},
year={2025},
organization={IEEE}
}

Review Summary

Chair Feedback

No comments provided.

🔖 Area Chair Comments

Review Group Report

Author Comments

This paper presents a very interesting approach to a real problem in autonomous driving, which is the difficulty of identifying materials that look similar in traditional RGB images.
The paper is of high quality and deserves to be presented at ICIP

Click here to view an online version of this email.

powered by Ex Ordo